

Presentation Deck

Slide 1

IntelliPlant AI

Industrial Knowledge Intelligence Platform

Transform Documents Into Industrial Intelligence

Slide 2

Problem Statement

- Fragmented information
 - 7-12 disconnected systems
 - Knowledge loss
 - Long search time
 - Repeated failures
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Slide 3

Our Solution

AI-powered Industrial Operations Brain

Using

- OCR
 - GraphRAG
 - Knowledge Graphs
 - Multi-Agent AI
 - Hybrid Search
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Slide 4

Architecture

Documents

↓

OCR

↓

Knowledge Graph

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Qdrant

↓

GraphRAG

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Multi-Agent System

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Gemini

↓

Answer

Slide 5

Innovation

✓ GraphRAG

✓ Multi-Agent System

✓ Knowledge Graph

✓ Explainable AI

✓ Compliance Intelligence

✓ Lessons Learned Engine

Slide 6

Multi-Agent Architecture

Supervisor Agent

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Expert Agent

RCA Agent

Compliance Agent

Lessons Agent

Slide 7

Knowledge Graph

Pump P101

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Bearing Failure

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Overheating

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Incident 2024

Slide 8

Demo Workflow

Upload Document

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OCR

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Graph Builder

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Ask Question

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Hybrid Search

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AI Answer

Slide 9

Business Impact

Reduce Search Time

30 min → 20 sec

Reduce Downtime

Improve Compliance

Knowledge Preservation

Slide 10

Technical Stack

Frontend

NextJS

Backend

FastAPI

AI

Gemini

LangGraph

Neo4j

Qdrant

Redis

Slide 11

Future Scope

Digital Twin

IoT

SCADA

Predictive Maintenance

Voice Copilot

Slide 12

Thank You

IntelliPlant AI

Industrial Operations Brain